

Payday Anomaly

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Abnormal returns have been found on days near the turn of the calendar months. Previous studies have linked the phenomenon to month-end paychecks, of which a sizable proportion goes into employees' retirement accounts and is then automatically invested in the market. Since many institutions adopt a semi-monthly pay schedule, we test the hypothesis that the market should exhibit detectable mid-month abnormal movement. Our results indicate that the 16th day of the month statistically and economically outperforms all other calendar days except the 1st and 2nd. As more and more institutions transition into bi-weekly pay schedule, however, the mid-month payday anomaly becomes less prominent.

The turn-of-the-month anomaly has important implications for academics and practitioners. For academics, the effect contradicts one of the founding principles of modern finance – that the market is at weak-form efficient. For practitioners, it presents a tradable pattern that investors can potentially take advantage of. Turn-of-the-month effect in equity returns has been well documented in existing literature. Lakonishok and Smidt (1988) first reported the anomaly by discovering that the four days at the turn of the month accounted for all the positive returns during the month with the remainder of the month generating a net negative cumulative return.

Some have linked the abnormal return to pay-day effect (Ogden, 1990; McConnell and Xu, 2008). Because employees get paid at the end of the month and many of them either automatically invest a portion of their paycheck in the market through retirement contribution or are encouraged to do so by having a surplus of funds with the new paycheck. But because many companies pay their employees twice a month, on the 15th and the end of the month, we hypothesize that on the condition that the pay-day effect holds true for the turn-of-the-month days, there should be a recognizable pattern in the middle of the month as well. After all, employees do make retirement contributions with every paycheck. To test this hypothesis, we analyze the S&P500 daily returns from January 3rd, 1950 to March 19th, 2018, the longest possible horizon from Yahoo Finance, the widely used public financial information source. We carried out a series of analyses to test whether there has been a paradigm shift in this payday anomaly.

Our results confirm that abnormal returns indeed exist in the middle of the month. The 16th of the month is not only the 3rd best day in the month overall but has moved up in the ranks monotonically every decade since the 1950s, until the most recent decade, the 2010s. We have limited evidence to believe that the increasing proportion of employees covered by defined-contribution plans such as the 401(k) contributed to the “rise” of the 16th day, and the recent trend in moving away from twice-a-month pay schedule to bi-weekly pay schedule has contributed to the decline of it.

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We conducted most of our analyses using the pivot table function in MS office. We believe this has significant implications to practitioners. By adopting a non-coding method, we demonstrate that practitioners can use the simple tool of pivot table, present in every office suite, to conduct comprehensive market data analysis.

Scheduled Investment

Savers typically schedule their investment contributions to coincide with their payday, where a portion of their gross income is allocated to investment. Examples of scheduled investment are commonly observed in 401(k), 403(b), and other cash-or-deferred arrangement (CODA) plans. With scheduled investment, professional asset managers as well as individual investors can expect recurring infusions of cash into asset markets that may induce asset prices higher. If the market is efficient, the recurring cash infusions into capital markets should be integrated into stock prices and should not yield any significant abnormal returns around those dates.

Calendar inefficiency is well documented in multiple instances such as the weekend effect, turn-of-the-month effect, January effect, holiday effect, and daylight savings effect. A multitude of studies document significantly higher stock returns on Fridays relative to lower returns on Mondays, commonly referred as the "weekend effect" or "turn-of -the-week" Effect. The weekend effect was first documented in Cross (1973) and results were replicated in French (1980) with over a hundred subsequent supporting studies. To our knowledge at least two studies have reported an abatement of the weekend effect in the latter end of the 1970s (Schwert, 2003; Robins & Smith, 2016).

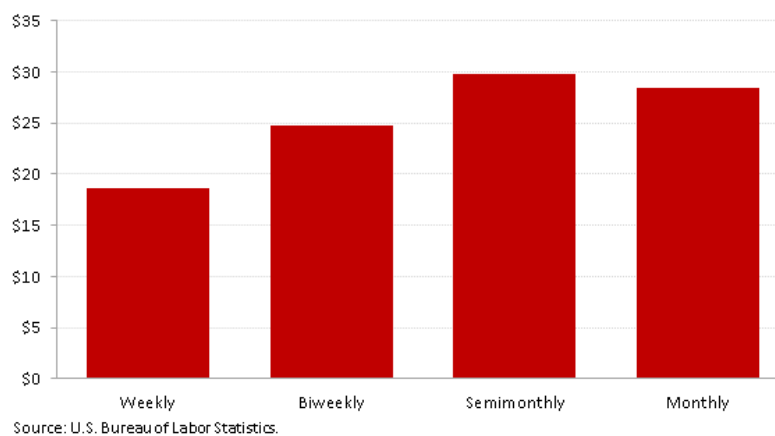
The "turn-of-the-month effect" considers monthly seasonality of above-normal returns in the last and first days of a month. As in the case of the weekend effect, numerous studies have documented abnormal returns. Ogden (1990) as well as McConnell and Xu (2008) suggest that the observed returns are likely attributable to cash inflows of "wages, dividends, interest and principal payments, and other liabilities." Both studies test for end/beginning of the month effect and do not analyze mid-month incidence. Ariel (1987) observed that positive returns only occur in the first half of the month. Similar first half of the month pattern was also found in end-of-week returns (Liano & Lindley, 1995). Gerlach (2007) found that macroeconomic news could explain for returns assigned to the turn of the month as well as seasonal weather.

Particular attention has been paid to the turn from December to January when assessing monthly seasonality. The transition from December to January, commonly known as the January Effect, identifies a consistent decline in end of December returns followed by an increase in January returns. Rozeff and Kinney (1976) document the January Effect over seventy years from 1904 to 1974 with Roll (1983) extended documentation of small firms over the period of 1962 to 1980. Employing ninety years of Dow Jones returns, Lakonishok and Smidt (1988) note that Dow Jones returns display the January Effect as well as weekend and turn-of-the-month effect. Reinganum (1983) and Ritter (1988) attribute the decline in end of December returns to investors closing positions in order to realize capital losses for the purpose of reducing tax liability. Many additional theories have been purposed to explain the January Effect such as Christmas bonus lock-in, window dressing, and market-cap concentration.

Who's Investing?

The Federal Reserve reported that during 1989-2013 the median defined-benefit and defined-contribution retirement wealth of the top half of wage earners is more than 150 times that of bottom wage earners (Devlin-Foltz, et al., 2016) - thus we should expect to see a larger flow of investment funds when top wage earners are paid. The Bureau of Labor Statics (2014) reports that the highest average hourly earnings are distributed semi-monthly followed by monthly distribution, see figure 1.² Since bi-monthly earners are paid mid-month and at the end of the month, these two reports indicate that we should focus this analysis on mid-month returns as the prior research has extensively documented a turn-of-the-month effect that has produced a number of competing theories. We test for the impact of scheduled investment tied to the mid-month paydays using daily returns of the S&P500 index over the period of 1950-2018.

Figure 1:
Average Hourly Earnings Paid by Private Businesses, by Length of Pay Period
March 2013



An Overall Look at the Payday Anomaly

Because practitioners have varying level of access to financial information, we decided to limit our study to only publicly available sources. We used the S&P500 adjusted daily close prices since January 3rd, 1950 to March 19th, 2018 to calculate its daily returns. Regular close price does not account for dividends and stock splits and therefore is not appropriate to calculate returns over a long-study horizon. Adjusted close price address this problem. Yahoo Finance is the only publicly available information source that offers time series of adjusted close price. Table 1 reports the mean return and its associated rank for each calendar day for the whole study horizon. Consistent with previous studies, three of the four turn-of-the-month days reported in Lakonishok and Smidt (1988), the 1st, the 2nd and the 31st were the best, the second best and the fourth best days in mean return. Not reported in any previous studies, the 16th day was the 3rd best day in mean return.

² In the United States, pay periods most commonly fall into four patterns: weekly, bi-weekly, semi-monthly, and monthly (Bureau of Labor Statistics, 2014).

Table 1: Mean and Rank of Calendar Day Returns						
Day of Month	Return	Rank		Day of Month	Return	Rank
1	0.180%	1		17	0.053%	11
2	0.148%	2		18	0.067%	9
3	0.101%	6		19	-0.130%	31
4	-0.005%	25		20	-0.023%	26
5	0.107%	5		21	0.022%	18
6	0.066%	10		22	-0.042%	28
7	-0.045%	29		23	-0.004%	24
8	0.004%	22		24	-0.003%	23
9	-0.072%	30		25	-0.029%	27
10	0.018%	20		26	0.045%	14
11	0.049%	12		27	0.014%	21
12	0.042%	15		28	0.077%	8
13	0.036%	17		29	0.080%	7
14	0.022%	19		30	0.045%	13
15	0.040%	16		31	0.118%	4
16	0.132%	3		Mean:	0.034%	

We term the four days, the 1st, 2nd, 16th and 31st, the pay days. We then tested whether the difference between returns for the pay days and those for the rest of the month is significant. Table 2, Panel A reports results from the t-test. We assume that the two series have different variances. For our study horizon, the mean return for the pay days is more than seven times the mean of the rest of the calendar month, and the difference is significant at 0.01. Table 2 extends the results of previous studies by including the 16th day of the month. Even though some previous studies also termed the effect the “payday effect”, none of those studies included the mid-month paydays (Ogden, 1990; McConnell and Xu, 2008).

A Detailed Look

For our study horizon, there have been some significant changes in how employees were paid in the U.S. In 1978, Congress passed The Revenue Act of 1978 with Section 401(k) paved the way for a new type of retirement benefit plans, the defined contribution plans. As the country slowly moved away from defined-benefit retirement plans to defined-contribution plans since the 1980s, more employees started participating in automatic retirement contributions, making the pay-period influx of funds into the market a prominent phenomenon (Davidson 2016). Table 2, Panel B reports t-test results covering a shorter horizon since the 1980s. The mean return for the pay days is now almost nine times the magnitude of that of the rest of the calendar month, and the result is significant at 0.01. The slight drop in statistical significance for the shorter horizon represented by the lower t stat is an apparent result of the much larger sample size for the whole study horizon.

Table 2: t-Test of Return Difference between Paydays and Other Days				
	A: 1950s to 2010s		B: 1980s to 2010s	
	Pay Days	Other Days	Pay Days	Other Days
#observations	748*	819	419**	459
Mean return	0.149%	0.020%	0.181%	0.022%
Hypothesized difference	0		0	
t Stat	5.10		4.06	
P(T<=t) two-tail	0.00		0.00	
Note: * For 71 months in the 1950s to 2010s period, none of the 1st, 2nd, 16th or 31st day was a trading day. We assume that returns for pay days and other days have different variances. ** For 40 months in the 1980s to 2010s period, none of the 1st, 2nd, 16th or 31st day was a trading day. We assume that returns for pay days and other days have different variances.				

We took a further look at the paradigm shift by generating a pivot table on a decade-by-decade basis. Table 3 reports that while the turn-of-the-month days of the 1st, 2nd and 31st all exhibited certain level of fluctuations in their rankings without any identifiable pattern, the 16th had consistently increased its position in the ranking until the most recent decade. It even became the best day of the month during the first decade of the 21st century. The result is intriguing to us because we do not believe that paychecks around the 16th would have been different from those received on the other pay days³. We suspect that because the other pay days have been extensively discussed in the literature, practitioners have been trying to take advantage of the anomaly and market efficiency has caught up, reducing the magnitude of the anomaly. We call for further studies into the phenomenon in search of an answer.

Table 3: Rank of Individual Pay Day, by Decade				
	Rank of 1st	Rank of 2nd	Rank of 16th	Rank of 31st
Overall	1	2	3	4
1950s	3	2	21	6
1960s	4	3	20	1
1970s	3	7	10	25
1980s	7	4	5	1
1990s	2	5	3	27
2000s	2	15	1	3
2010s	8	4	3	30

³ These checks are likely received and sent to the financial institutions on the 15th of the month, which were then invested in the market place on the following day, the 16th.

We have limited evidence to believe that the drop in the ranking for the 16th day in the current decade is partially because many U.S. companies are moving away from the semi-monthly pay schedule to a bi-weekly pay schedule. The advantage of the latter over the former is standardized compliance. Federal law does not require or specify a frequency of pay, whereas most U.S. states do. More than two-thirds of states require that paydays occur no less frequently than monthly or semi-monthly. However, employee payment is complicated by the fact many of the states with monthly or semi-monthly payment laws require more frequently occurring paydays for different occupation types. For business having employees in multiple states and, or of various occupations, compliance is simplified by adopting the bi-weekly pay schedule. See Figure 2.

Figure 2:
Percentage of Private Businesses Operating under
Weekly, Biweekly, Semi-monthly, and Monthly Pay Periods
March 2013

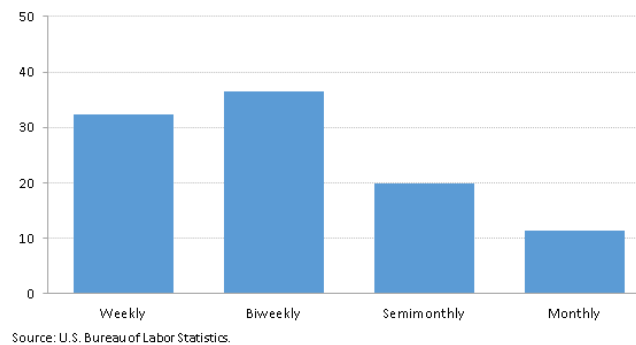


Table 4 isolates the 16th day against other days in the month. Panel A reports the result for the whole study horizon of 1950s to the 2010s, while Panel B reports the result for the shorter horizon of 1980s to the 2010s. During both time frames, the difference between the mean return for the 16th and that of the rest of the month is significant, both economically and statistically. Since the 1980s when more and more employees started making automatic retirement contribution (Davidson 2016), the difference has been much more prominent, and the statistical power went up despite a smaller sample size.

Table 4:				
t-Test of Return Difference between the 16th and Other Days				
	A: 1950s to 2010s		B: 1980s to 2010s	
	16th	Other Days	16th	Other Days
#observations	572*	819	320**	459
Mean return	0.132%	0.031%	0.214%	0.033%
Hypothesized difference	0		0	
t Stat	2.62		3.07	
P(T<=t) two-tail	0.01		0.00	
Note:				
* For 247 months in our study horizon, the 16th was not a trading day. We assume that returns for the 16th and other days have different variances.				
** For 139 months since January 1980, the 16th was not a trading day. We assume that returns for the 16th and other days have different variances.				

Further Discussion

In our analysis, we assume that the semi-monthly paychecks are distributed on the 15th of each month, with the retirement contribution portion reaches the financial institution at the end of the day and ready for investing into the market on the following day, the 16th. Another assumption we made is that a significant portion of the retirement contribution funds were invested in broad-market funds, such as the S&P500 fund. That is the rationale for using the S&P500 index to study the payday anomaly. Further studies in these areas will provide more insight into this phenomenon.

If there is a causal relationship between semi-monthly paycheck deduction and abnormal return observed on the 16th of the month, would any major economic event, e.g. a dramatic increase in unemployment, have a measurable impact on the magnitude of the return anomaly on the 16th during the same period? In future studies, we would like to explore further into intricacies of such a relationship so that we may determine whether this relationship is causal in nature.

Conclusion

In the current study, we expand the study on payday anomaly to include mid-month days. We found that the 16th day systematically outperforms the other calendar days, except two other paydays, the 1st and the 2nd days of the month. This confirms our hypothesis that semi-monthly pay system contributes to detectable abnormal returns not only at the turn of the month, but in the middle of the month as well. Our result is consistent with a trend in the US retirement system since the 1980s – more companies moving away from defined-benefit plans and moving into defined-contribution plans. As bi-weekly pay system slows gains more popularity over semi-monthly pay system, however, we have noticed a reduced prominence in the mid-month payday anomaly.

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